

Mohamed Abdelkader

Software Engineer — Full Stack & DevOps Explorer

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Professional Summary

Adaptable Software Engineer with an **Engineering Mindset**, specializing in the MEAN stack. Currently expanding expertise in **Cloud Operations and DevOps (DEPI Track)** to build scalable, high-availability systems. Possess a strong foundation in **Data Analysis and Machine Learning**, enabling a data-driven approach to software architecture and system optimization. Proven ability to integrate AI tools and optimize backend performance.

Education

Benha University, Faculty of Computers and Artificial Intelligence

Bachelor of Science in Information Systems

Qalyubia, Egypt

Oct 2022 – Jun 2026

Experience

MEAN Stack Web Developer Trainee

National Telecommunication Institute (NTI)

Jun 2025 – Sep 2025

Benha, Egypt

- Completed a 210-hour intensive program, mastering Node.js, Angular, and MongoDB to build secure, scalable RESTful APIs and modern SPAs.
- Implemented JWT-based authentication and Role-Based Access Control (RBAC) to ensure enterprise-level security.
- Optimized database queries and schemas, improving data retrieval efficiency for high-traffic applications.

Technical Skills

Languages: JavaScript, TypeScript, Python, SQL (PostgreSQL/MySQL), C++, Java, PHP

Backend: Node.js, Express.js, RESTful APIs, **Redis (Caching)**, Socket.io, JWT, Mongoose ODM

Frontend: Angular, React (Basics), RxJS, NgRx, Bootstrap 5, Responsive Design

Data & ML: Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Exploratory Data Analysis (EDA)

DevOps & Tools: Git, Docker & Kubernetes (Learning), Linux, Postman, AI-Driven Development (Gemini/Claude)

CS Fundamentals: Software Architecture, Data Structures, Algorithms, OOP, Database Design (NoSQL/RDBMS)

Projects

3yadetak (Healthcare Management System) | Node.js, Flutter, MongoDB

Current

- Leading the development of a complex healthcare system as a graduation project, focusing on secure medical record management and real-time scheduling.
- Designing the system using **Clean Architecture** principles to ensure scalability and ease of maintenance.

Diabetes Risk Prediction System | Python, Scikit-learn, ML Models

2025

- Developed a machine learning pipeline comparing 5 models (SVM, Random Forest, etc.) to predict diabetes risk using a dataset of 250K+ samples.
- Performed extensive EDA and feature engineering, identifying key health indicators and handling class imbalance issues.

Audio Library & Streaming Platform | Node.js, Express.js, MongoDB, Socket.io

2025

- Engineered a high-availability audio streaming backend supporting concurrent users with robust file upload systems using Multer.
- Integrated social features including personalized feeds and interaction systems, boosting user engagement metrics.

YouTube Clone Platform | MEAN Stack, Video Streaming, Redis

2025

- Developing a video platform with chunk-based streaming and real-time notifications via Socket.io.
- Implementing **Redis** for caching metadata to minimize database load and improve response times.

Certifications

MEAN Stack Career Accelerator NTI Certificate Link • 210-hour intensive program covering full-stack development and freelancing business skills.	2025
National Certified Freelancer ID ITIDA Certificate Link • Officially certified freelancer by the Information Technology Industry Development Agency (ITIDA).	2026

Languages

Arabic: Native
English: Professional Working Proficiency